

IT462 - EDA

Tools

-

An

Open

Closed

MIND

HEART

- Python

-

(Google Colab)

numpy

Pandas

17, 41, 30, 99

~~data~~

~~Scores with AVG~~
~~info 50~~

Data

raw, unorganized
unstructured

Structured

information

Text

Audio

Signals

Records

Source

Primary Data (I am involved)

Secondary Data (You are involved)

Tertiary Data

Data

File
formats

• CSV ✓

• XLSX ✓

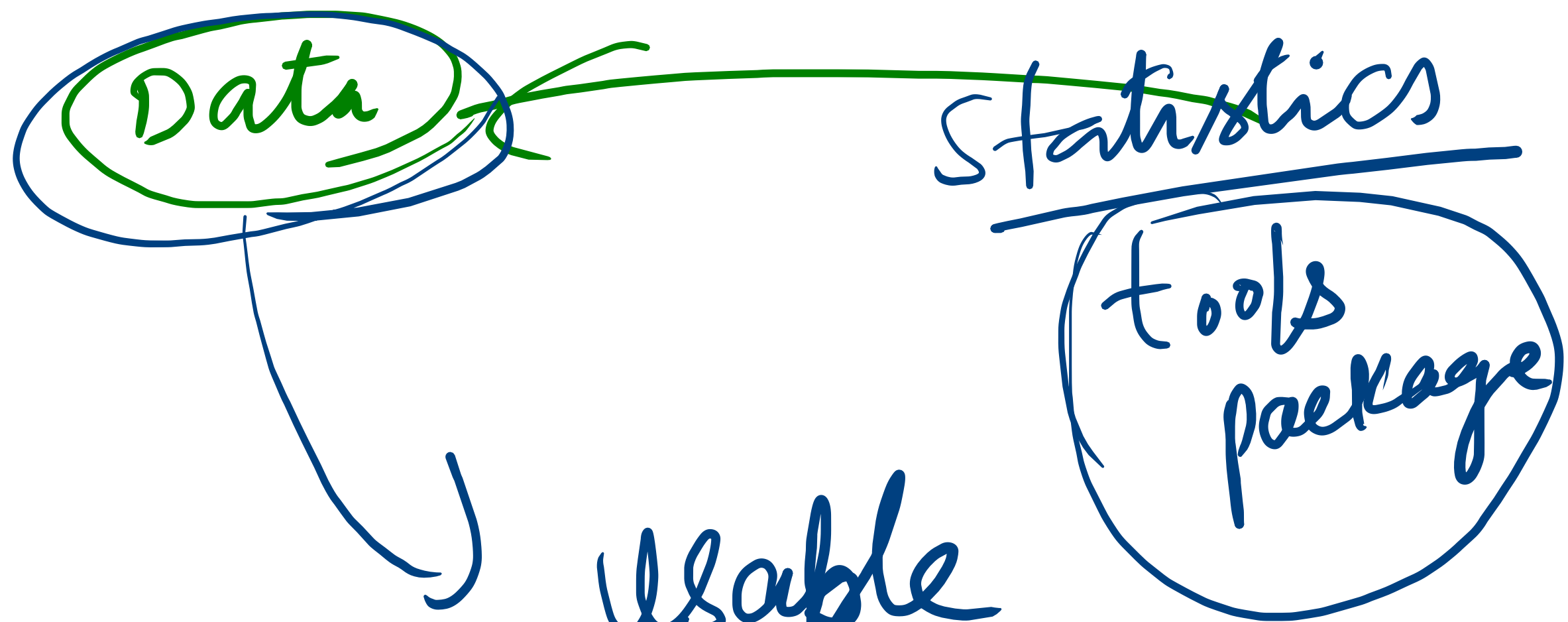
• json

• html

~~Web scraping~~

Python

text



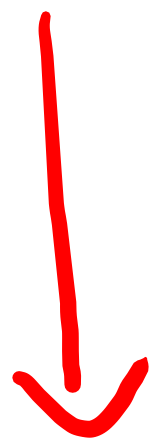
IDA
Initial
Data

Analysis

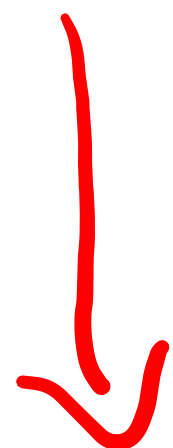
- validity of
- reliability
- authenticity
- error free

ID	IT462	Age _y	Gender
2026.	10	0	M
2026	50	10	F
.	90	20	M
.	-20	100	F
2023	17	:	-

IDA



EDA



CDA



features of Data



modelling
Evaluation



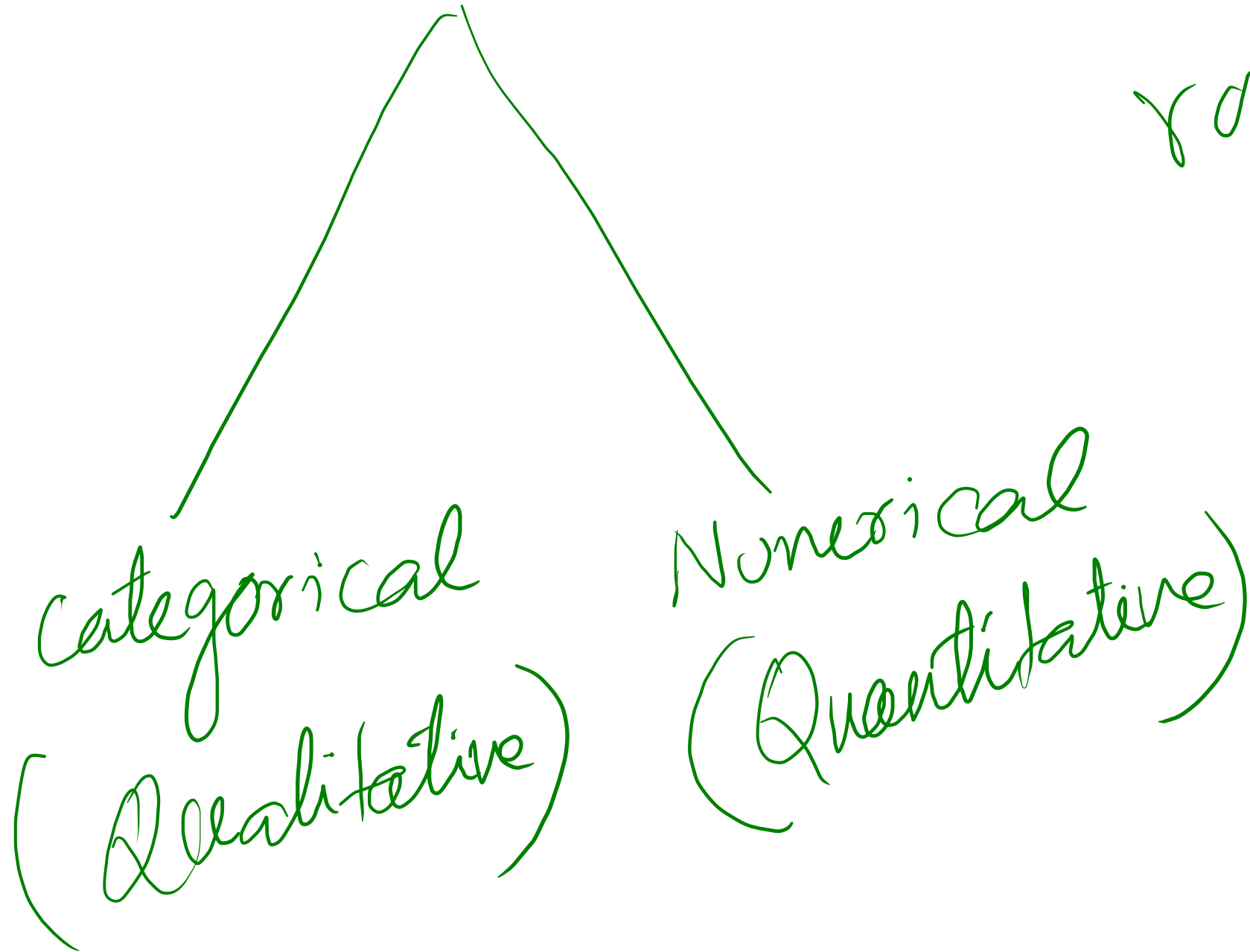
Structured Data

Columns → Variables
or
features

rows →

observations

or
values



Measurement Scales

Categorical

Nominal —

Ordinal

Names

Indexes

Tags

~~Patents~~

~~Color~~

Nominal

~~Size~~

Ordinal

Nominal

Interval

Ratio

N
-
O
-
R

Measurement Scales

① Nominal

② Ordinal

③ Interval

④ Ratio

} Category →

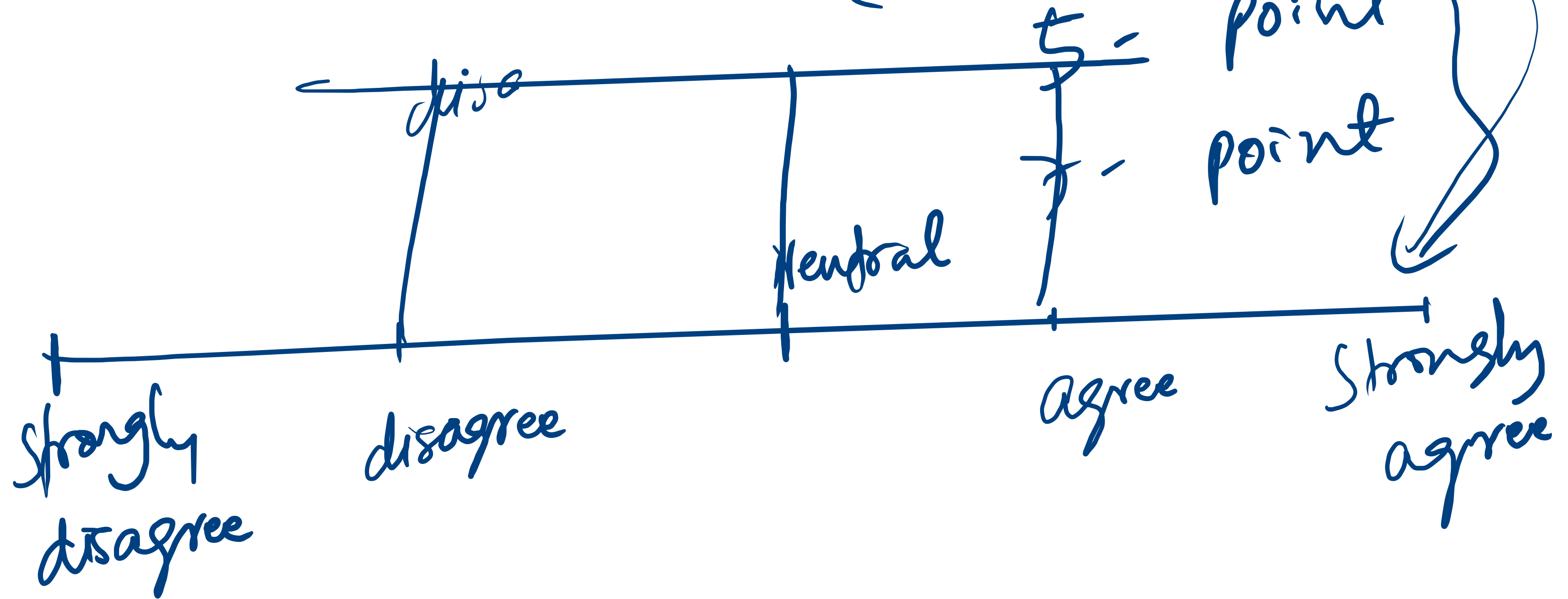
Labels
category

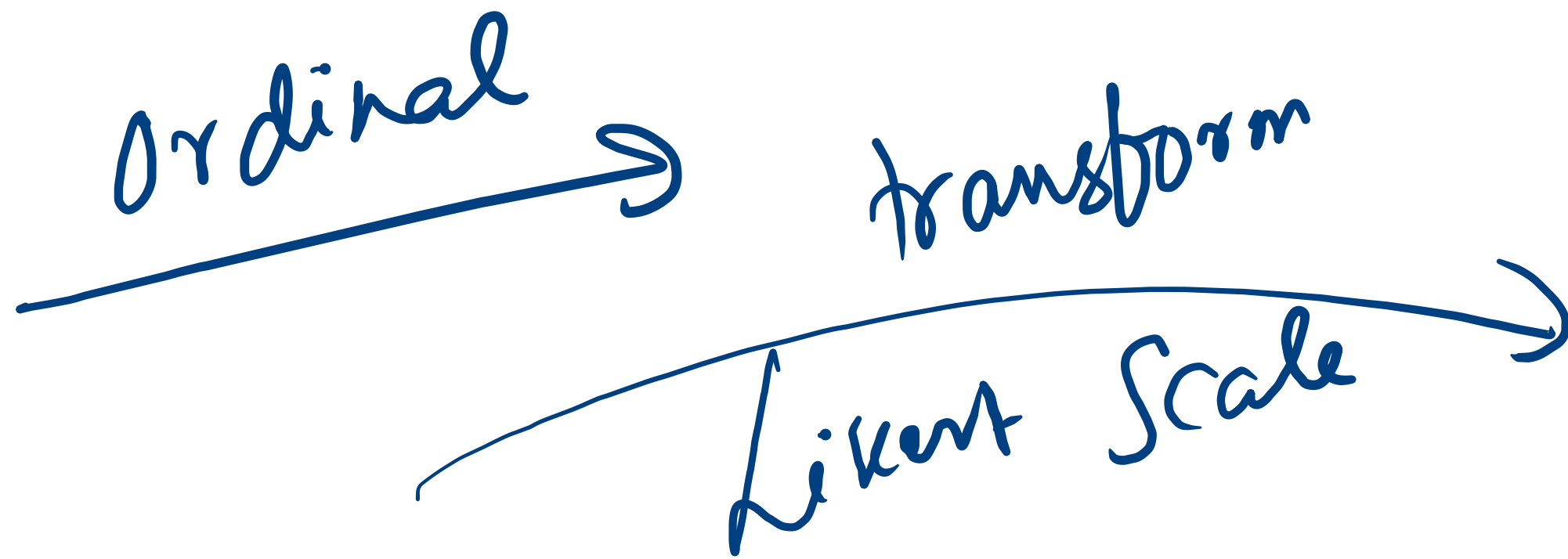
Numeric data

quantities

<u>Data</u>	N	O	I	R
<u>Labeled</u> date	✓	✓	✓	✓
inherent order	x	✓	✓	✓
measurable difference	x	x	✓	✓
Absolute Zero (True Zero)	x	x	x	✓

Likert scale →





- Interval
- make quantifiable equivalence with numbers
 - make the difference equal in Length

Age

Ratio

(because of Absolute Zero)
Person with age 0 is impossible)

(A

Variables

features

///

} values

Operations

N	O	I	R
X	X	+ -	X / o

Dataset variables measurement scales

% , freq , counts
prop

rank

Step 1

Step 2

Step 3

Ratio

+ , -

x , ÷

yes

Step 4

no

Can I

Can I

Can I

Can I

I

Y

Y

Y

Y

Yes

Count

Nominal rank

yes Ordinal

measure the difference?

Can I find the ratio?

Identity Group!

X

no

Interval

Statistical tools

Descriptive

Inferential

→ Data

Summary

Description

different

measures

→ Central tendency

Mean
Median
Mode

Statistical measures for measurement scales

(Completed)

① Measures of central tendency:

Mean

- Arithmetic
- Geom ()
- Harmonic mean

Median

Mode

Trimmed - ~~mean~~ ^{mean} weighted after removing the extremes: $x_i, i > 1 \dots n$

$$\frac{\sum w_i x_i}{\sum w_i}$$

1 2 4 6

	sort	exp	variants
Mean	x	$\sum \frac{x_i}{n}$	Trimmed - mean ^{mean} weighted after removing the extremes: $x_i, i > 1 \dots n$
Median	✓	$\uparrow$ n-odd $x_{(\frac{n+1}{2})}$ n-even	
Mode	✓	$\frac{1}{2} (x_{\frac{n}{2}} + x_{\frac{n}{2}+1})$ median	

②

Measures of Dispersion / Variation

$x_1, \dots, x_n$

Range: —

$$x_H - x_L$$

x_H — max^m Score

Variance —

popⁿ

$$\sum_{i=1}^N (x_i - \bar{x})^2$$

x_L — min^m Score

Standard deviation —

Sample —

$$\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$$

variance

Quartile Deviation

Percentiles

Quartile
quantile

decile :-

Percentile position of x_i

$P(x_i)$

$$= \frac{(i-1) \times 100}{(n-1)}$$

$$\frac{i}{n} \times 100$$

2, 3, 5, 6, 8, 10, 12, 15, 18, 20

_{1 2 3 4 5 6 7}

$n = 10$

$x_i = 12$

$P(x_i)$

$$P(x_i) = \frac{i-1}{n-1} \times 100 =$$

$$\frac{7-1}{10-1} \times 100 = \frac{6}{9} \times 100 = 67\%$$

Q1

What is

Q2:

Find

the 25th percentile

Percentile

$$P \times \frac{(n-1)}{100} =$$

$$\frac{25 \times 9}{100}$$

$$\frac{9}{4} = 2.25$$

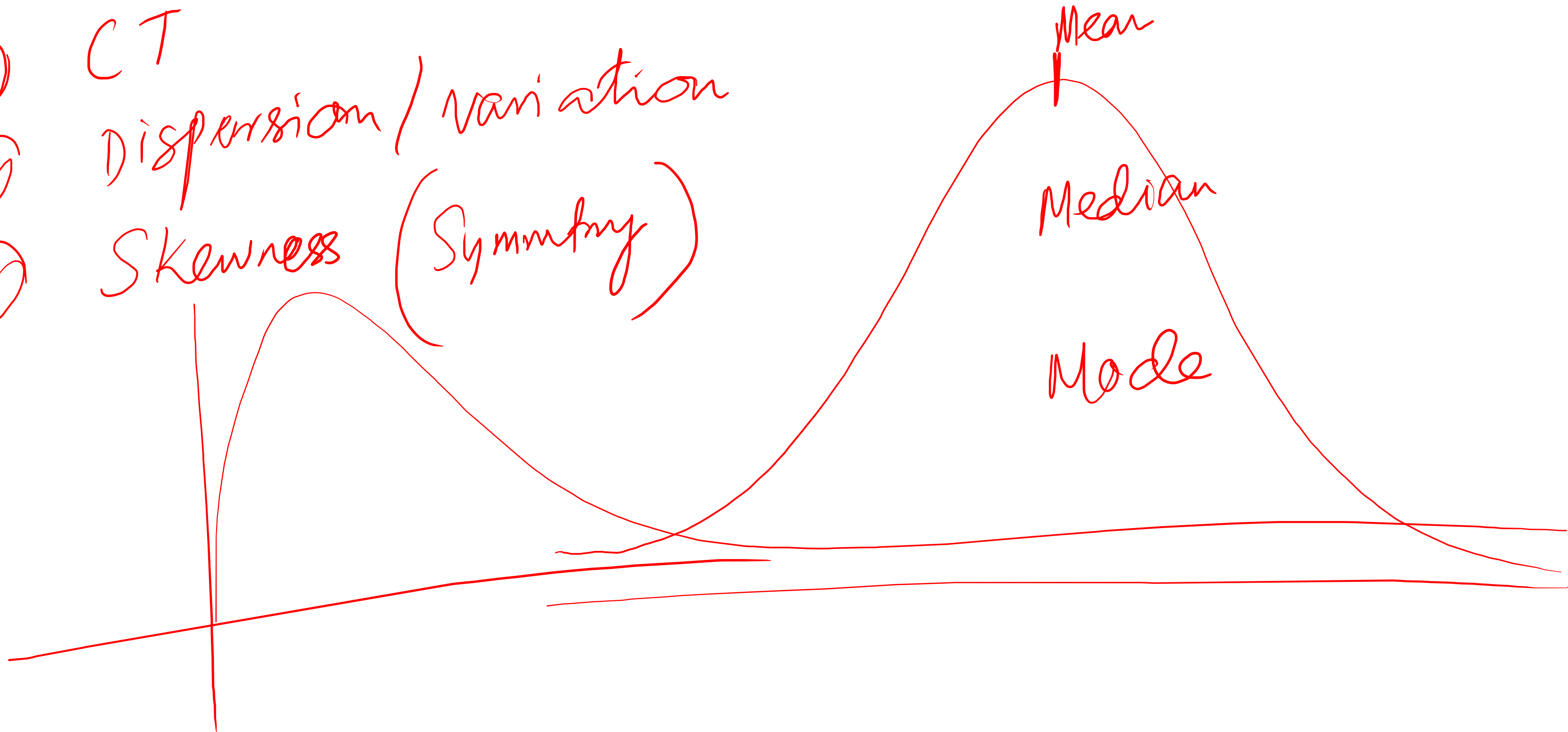
$i = 1 = 2.25$
 $i = 3.25$

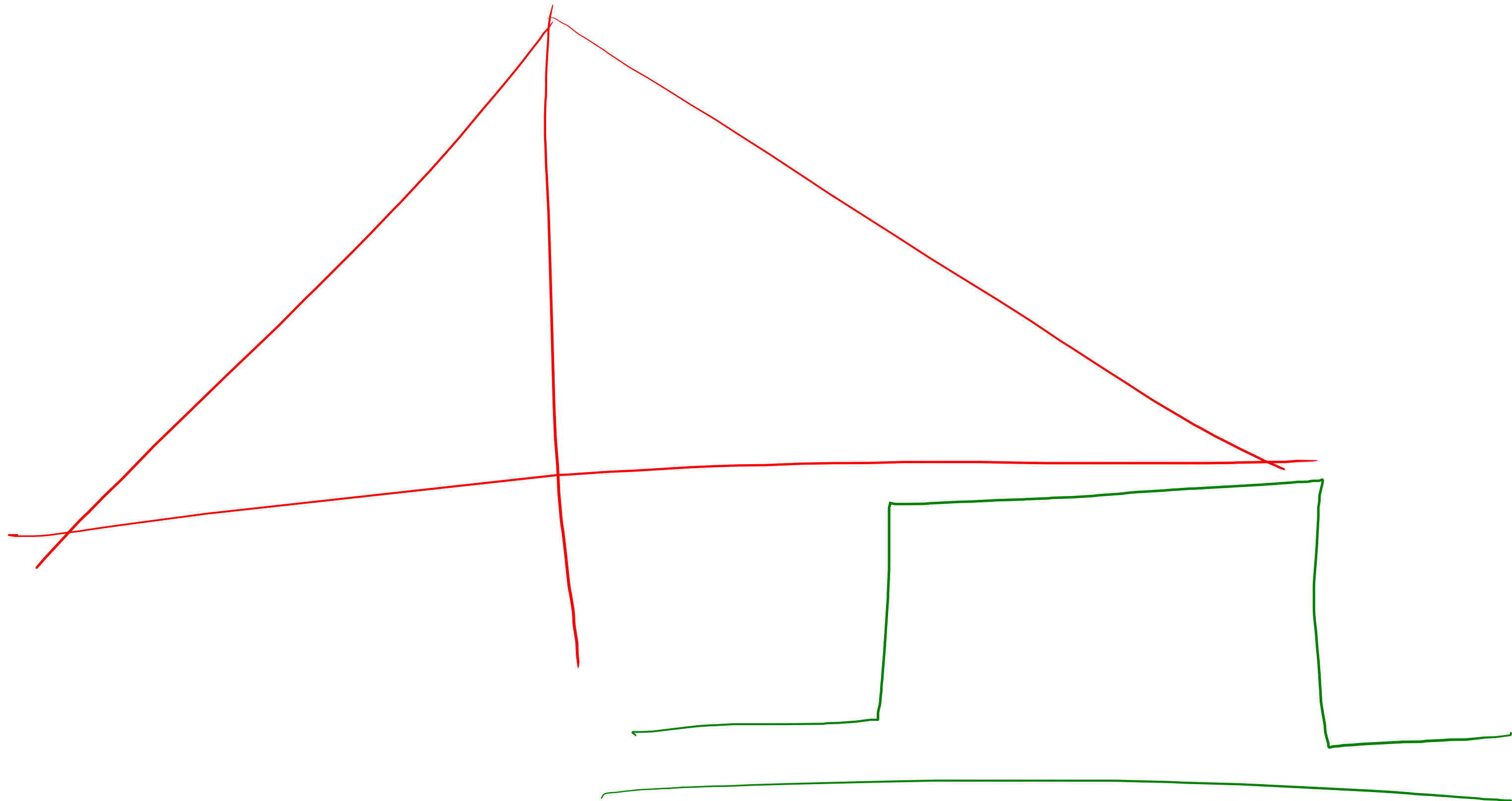
$$i-1 = \frac{P(x_i) \times (n-1)}{100} =$$

$i-1$

Statistical measures

- ① CT
- ② Dispersion / variation
- ③ Skewness (Symmetry)





Skew function ← Skewness

df

Pandas

df[''].skew()

④

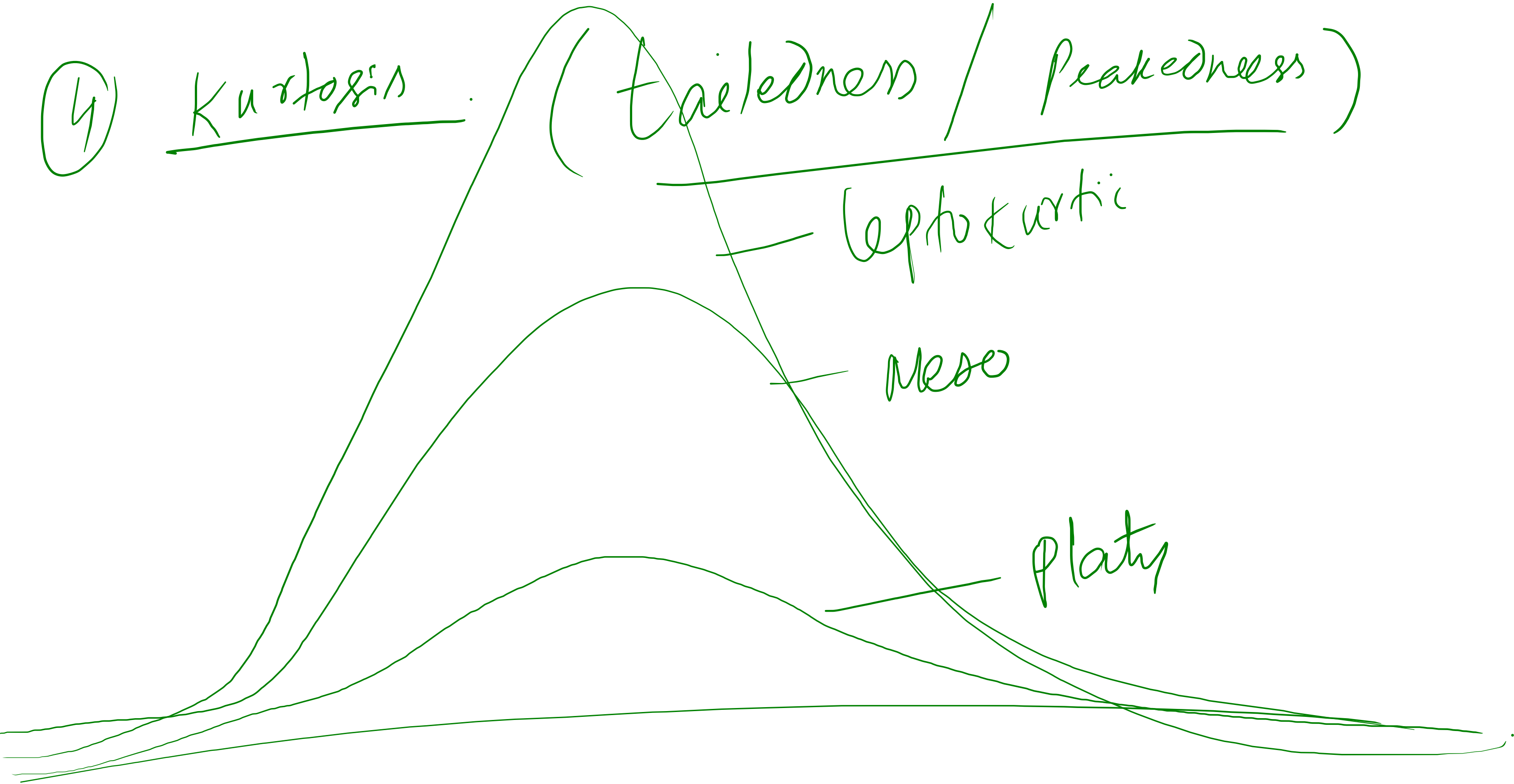
Kurtosis

(tailedness / peakedness)

leptokurtic

meso

platy



⑤

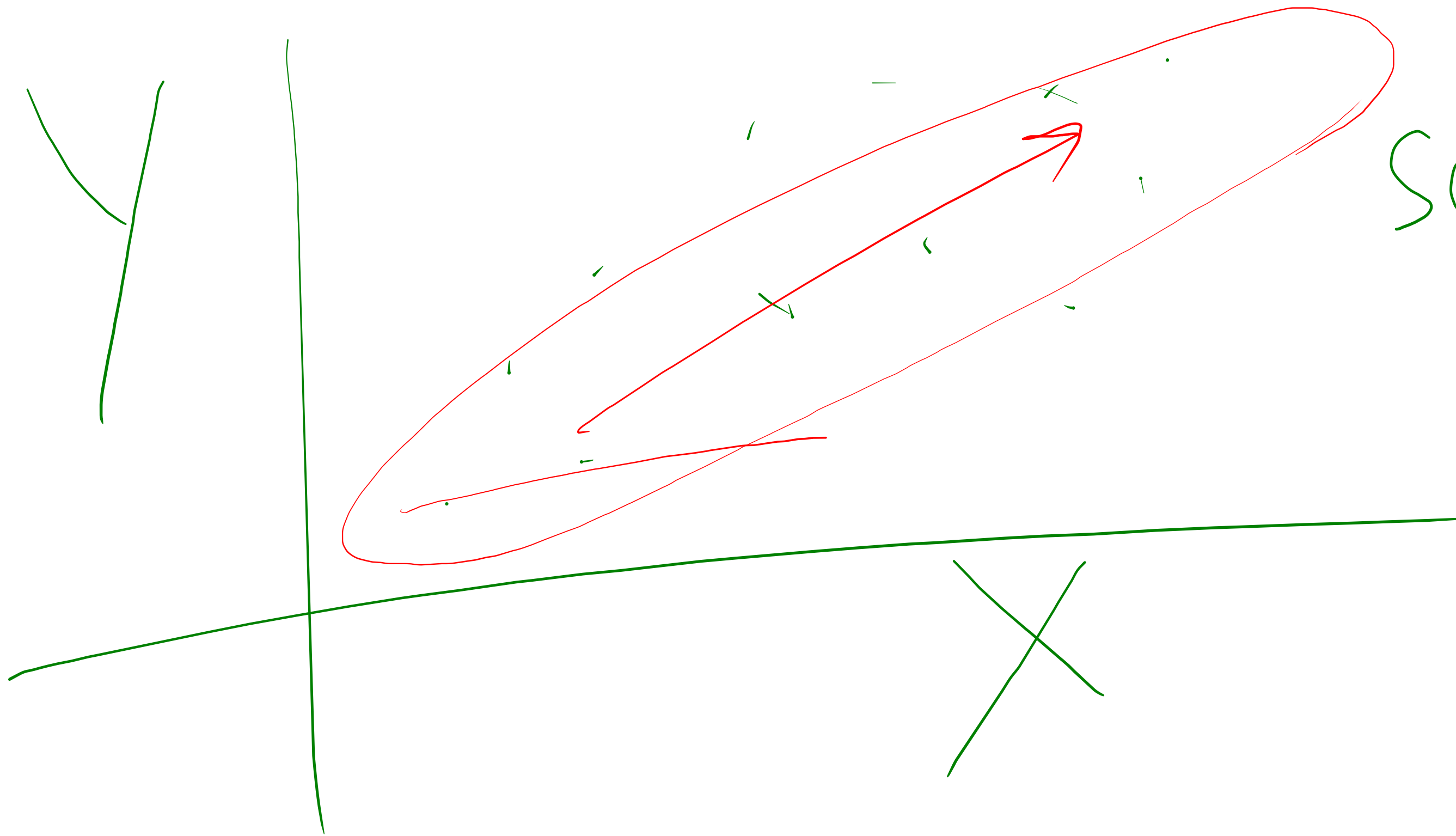
Measures of relationship / association

variables $\rightarrow X, Y$

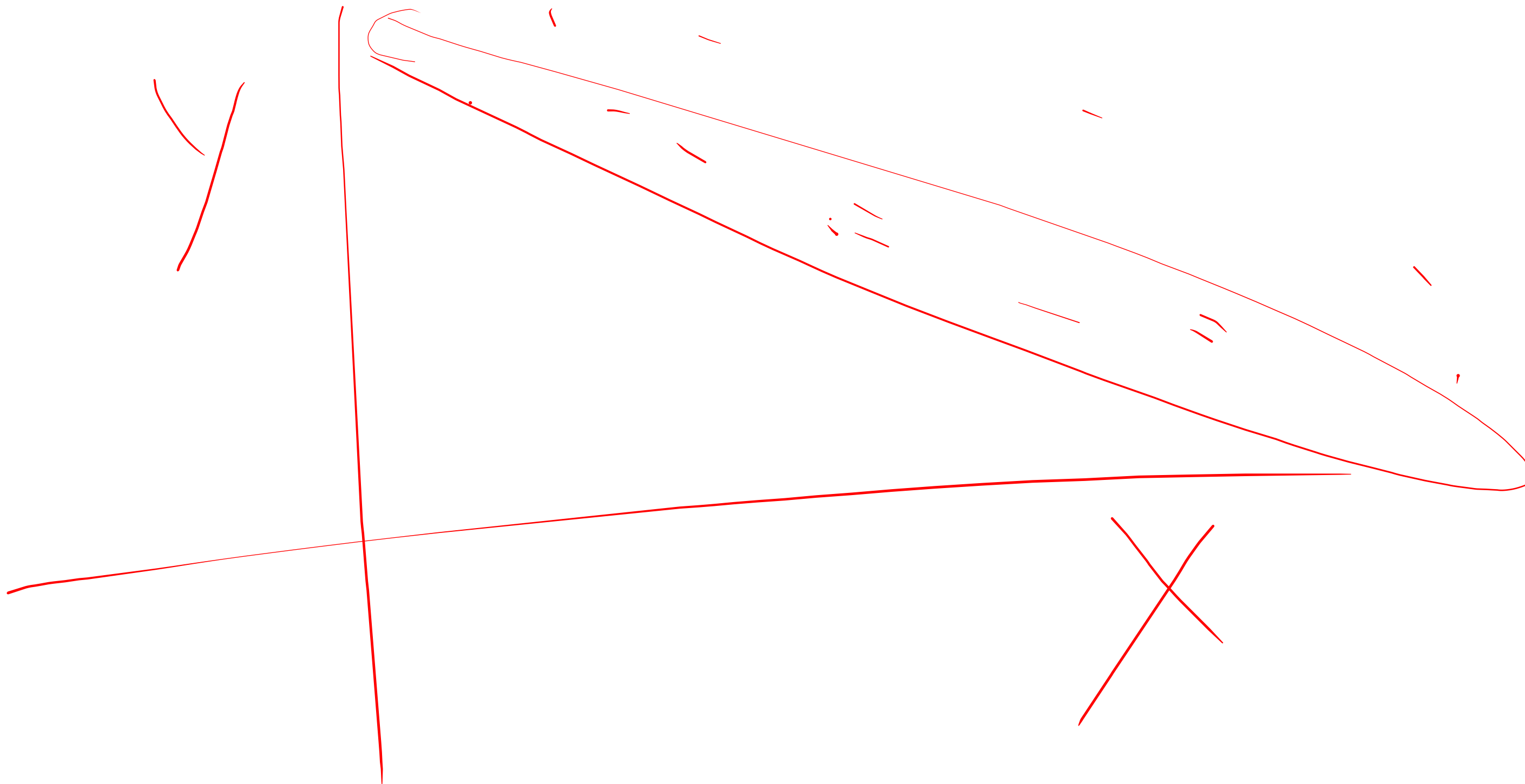
Linear relationship



Non-linear relationship

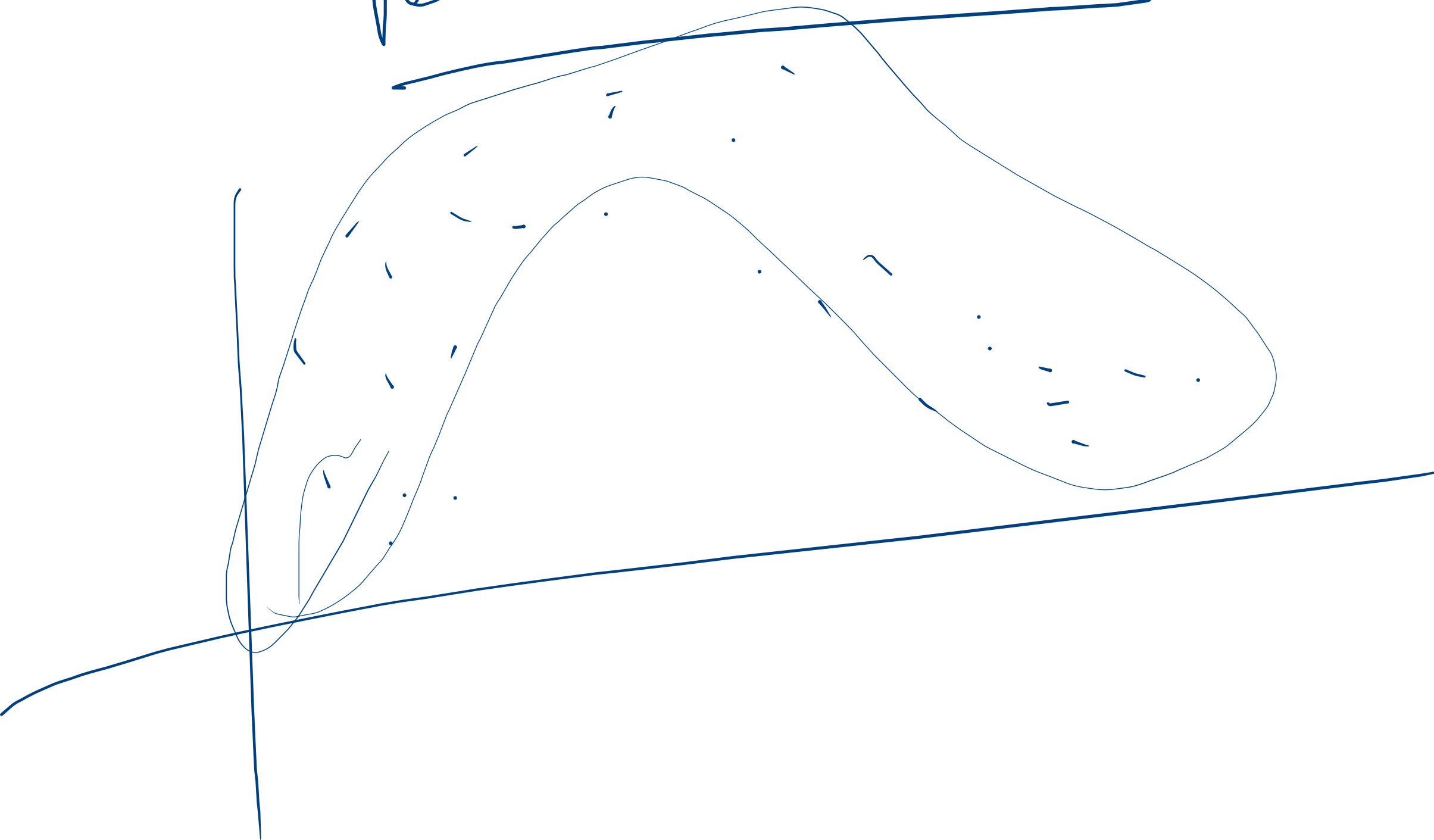


Scatter
plot



Pearson Correlation

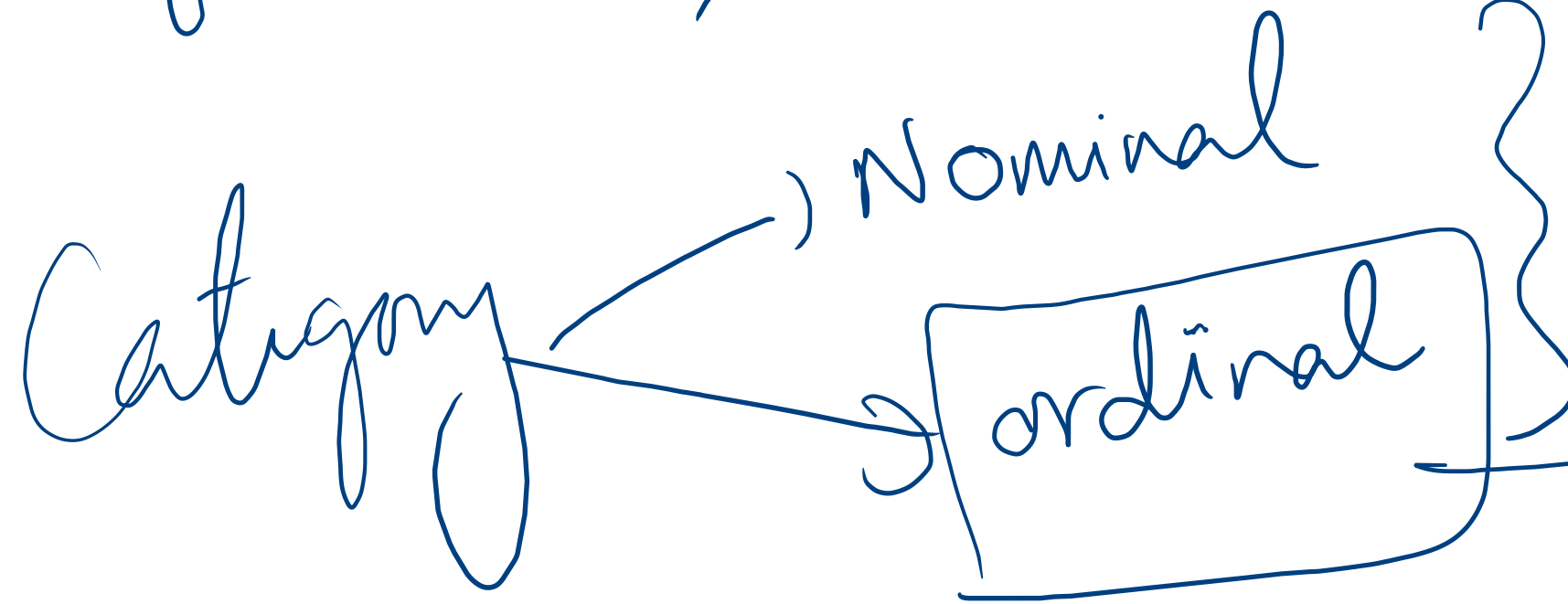
df. $\text{corr}()$



Interval
& ratio



Pearson r



Spearman r_s

Y - cts

binomial

X - binomial

binomial

point biserial
Correlation

Cramer ϕ

scipy. stats — ss

Spearman $r \rightarrow$ ss. spearmanr(x, y)

pointbiserial $\rightarrow$ ss. pointbiserialr(x, y)

Cramer's $\phi / V \rightarrow$

Cross tab

Contingency table

in statsmodels

target

Data →

↓
characterize
(measure scales)

Statistical tools

Expt

eg Effect
of
Vaccination

↓
Sampling

Indian
population

n size

Sample.

↓
each unit/entity
observed.

↓
Statistic

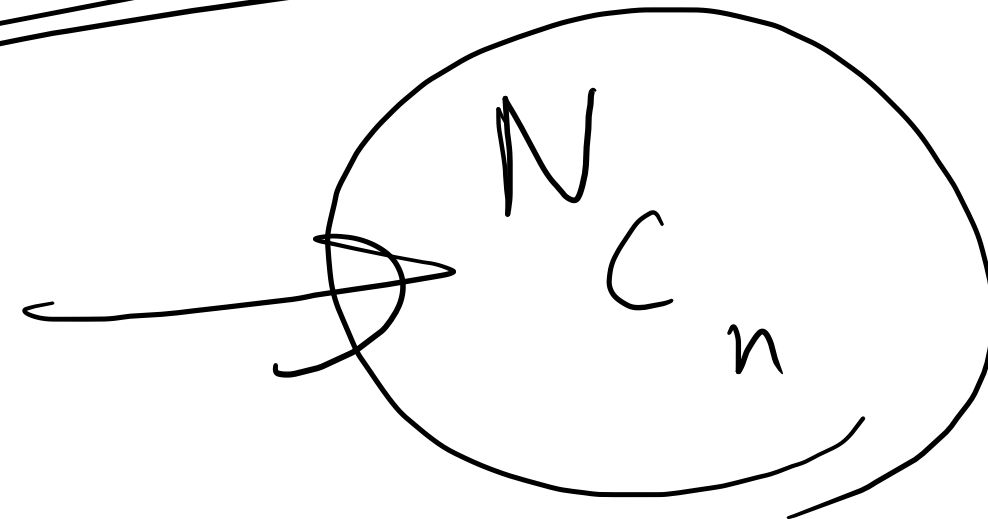
N
size

population

↓
parameter

$N \gg n$

Who should I observe?



methods of sampling

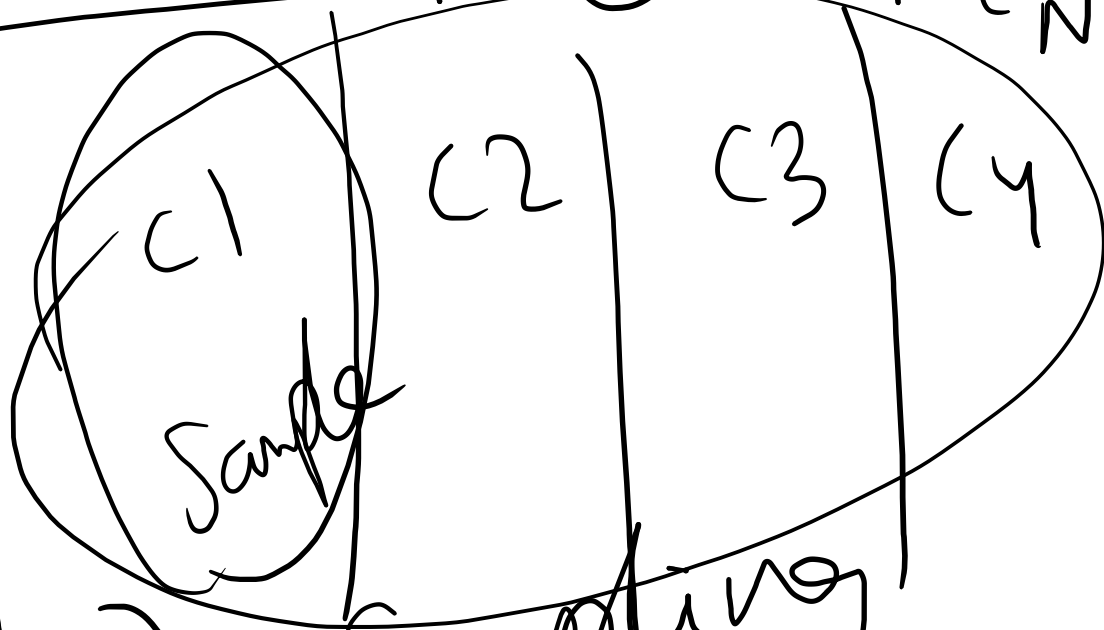
- Simple Random Sampling

N
 C_N

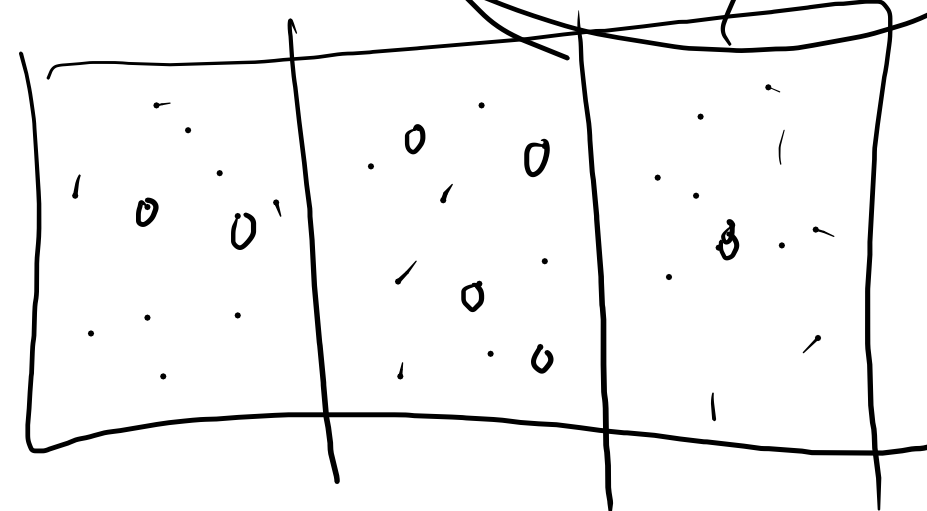
1, 2, 3, 4

Cluster Sampling

N
 C_N



- Stratified Sampling



2 points

1, 2
1, 3
1, 4

Expt Design

Sampling Bias



Problem Defⁿ

Design of Expts

Research Question

target

Response var

Experimental unit

Factors / levels

Choose Design



Randomization Replication control/Block

Collect data

↓
IDA

↓
EDA

Is the Data reliable?

What patterns are there?

Data Input

(CSV)

①

features



①

dimension — df.shape

②

type (measurement scales)

③

scaling Convention

④

Units, date-time (values)

⑤

duplication

⑥

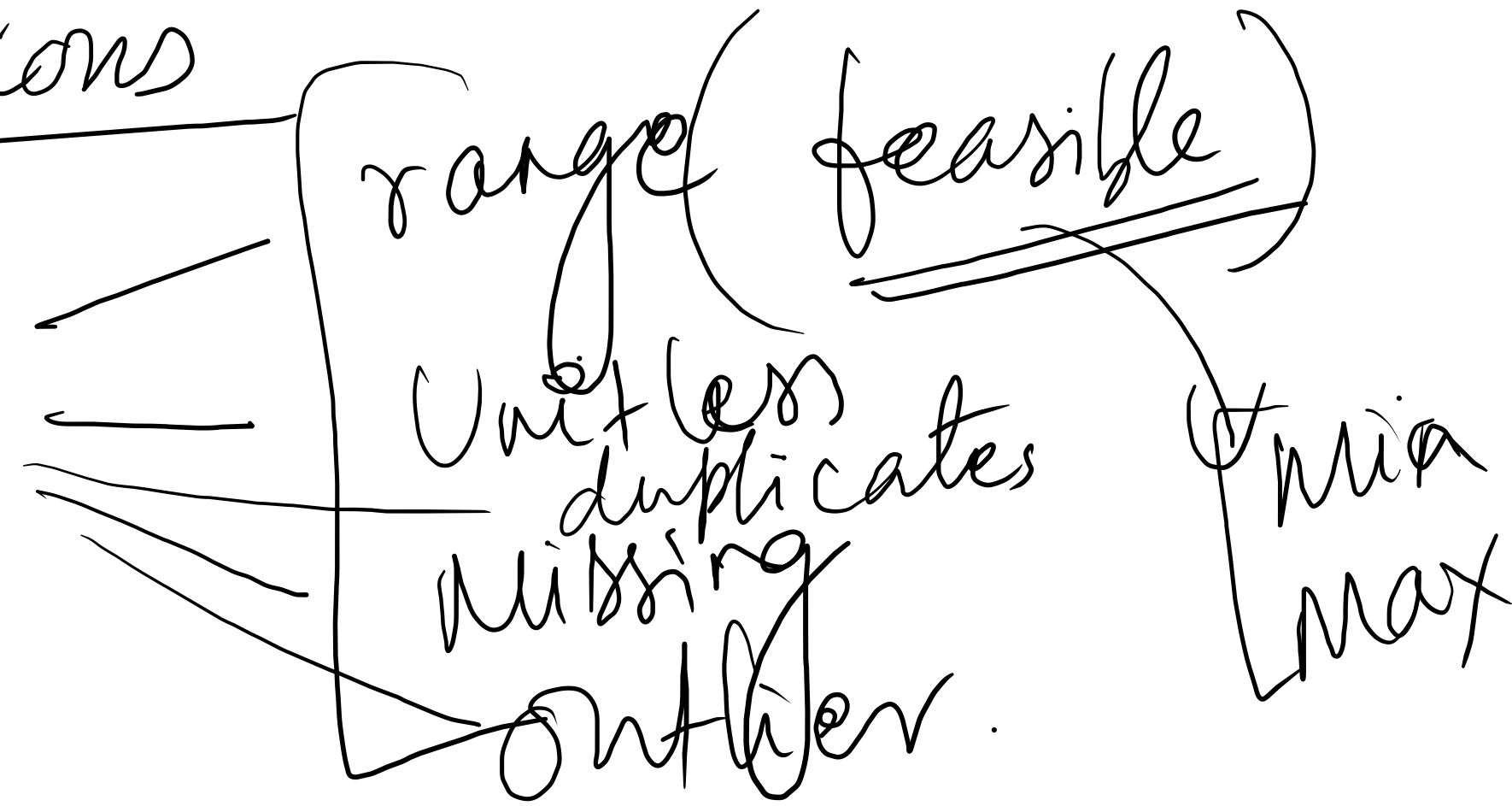
..... add on

⊕

feature Engineering ✓

②

Values / observations

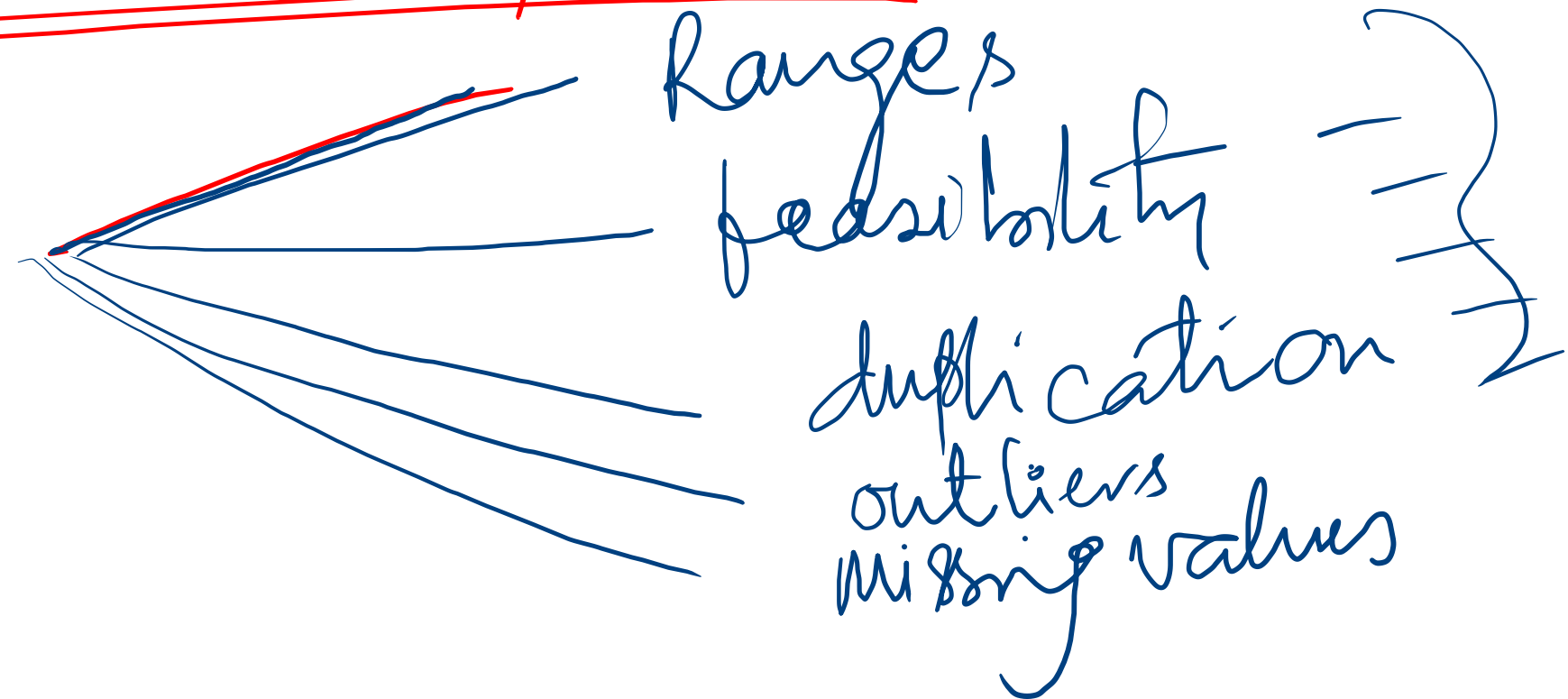
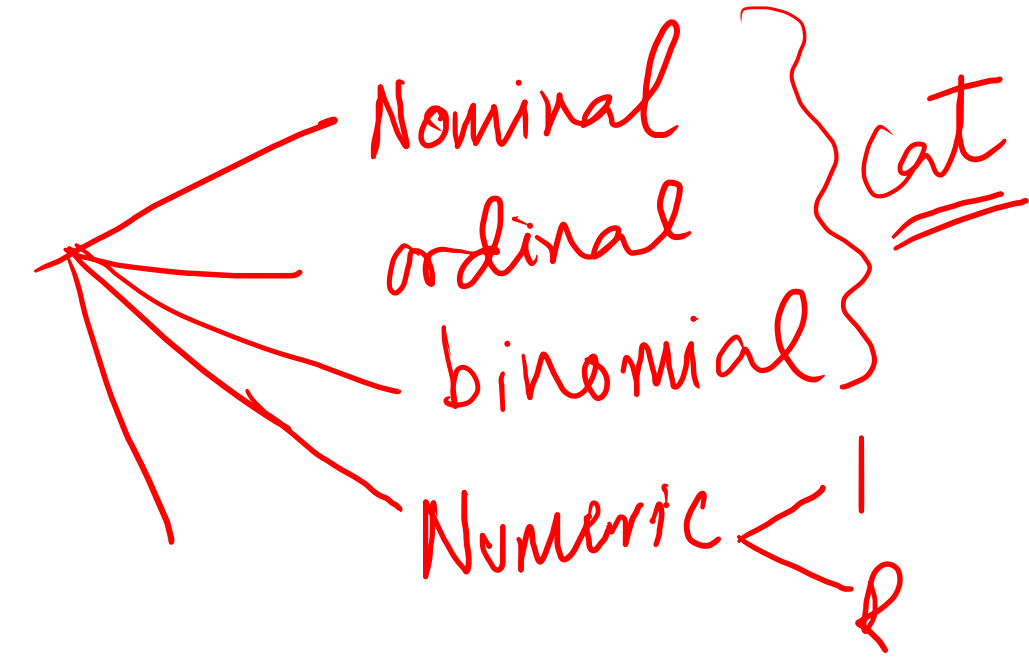


✓ Imported Data

CSV

✓ features / Columns / variables

✓ values / observations / records



Definition

Outliers $\rightarrow$ values / obs that
lie distant from most
of the obs

$\rightarrow$ extreme values

Detection

Numerical $\rightarrow$ IQR

$x \notin [Q_1 - 1.5 IQR, Q_3 + 1.5 IQR]$

Graphical



IQR

$1.5 IQR$

$X \notin \underline{\underline{\pm 3z}}$

②

z-score

Graphical method

①

Histogram



②

Box plot

